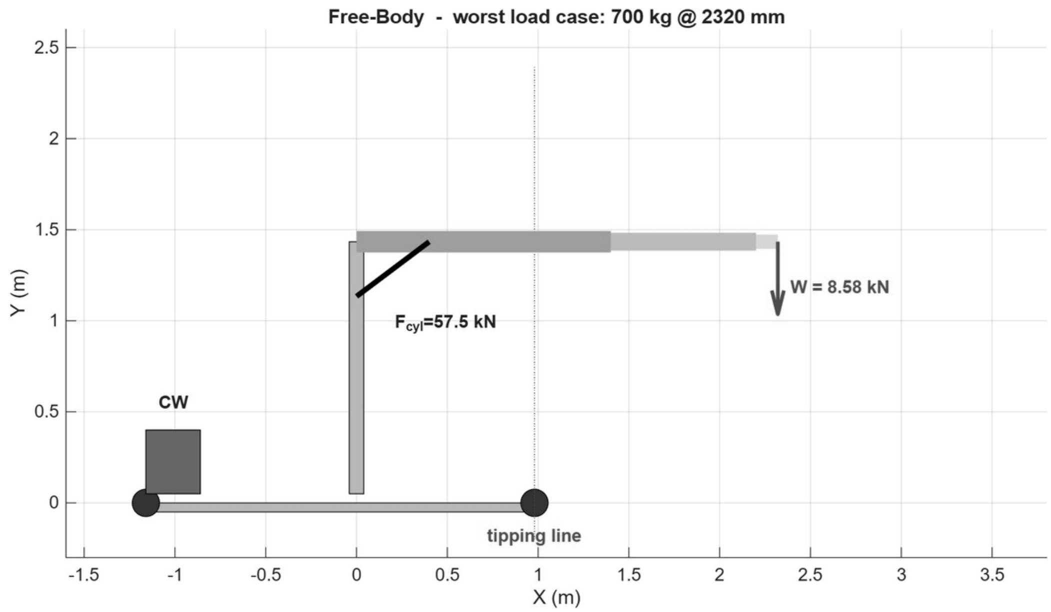
**PROJECT 40467361**

# STRUCTURAL CALCULATION AND LOAD VERIFICATION REPORT

## 1 Tonne Mobile Floor Jib Crane

*Combined verification: wheels-only and deployed-outrigger configurations*




*Free body at the governing case, 700 kg at 2320 mm. The tipping line is at the front wheel, 980 mm ahead of the column.*

| Field | Detail |
|---|---|
| Project reference | 40467361 |
| Configurations assessed | Outriggers stowed (wheels only) and outriggers deployed (# arrangement) |
| Rated capacity (as marked) | 1000 kg at 1400 mm reach |
| Document revision | Rev 3 (combined; supersedes wheels-only Rev 2 and deployed Rev A as a single submission) |
| Date | 10 June 2026 |
| Status | Issued for review |
| Analysis tool | MATLAB R2026a (parametric verification model, project 40467361) |
| Companion document | Load Re-Rating Report (Rev R1), issued separately |

# Document Control

This report is the combined independent verification of the 1 tonne mobile jib crane shown on the manufacturer drawing set (Assembly, Base, Arm Section 1, Arm Section 2). It assesses the crane in both of its working stances in a single document: standing on its four wheels with the outrigger legs stowed, and standing on its four deployed legs in the # arrangement. It draws together the previously separate wheels-only Rev 2 report and deployed-outrigger Rev A report, which share one parametric model and one set of inputs, so that the full picture is available to the reviewing officer in one place.

The two configurations differ only below the slewing bearing, in stability and in the members that carry the load to the ground. Everything above the bearing, the column, the boom, the pins, the welds and the slewing bolts, is fixed by the boom geometry and the lifted load and is identical in both stances. The report is organised to make that division explicit: a common basis, then each configuration, then a combined outcome.

## Revision history

| Rev | Date | Description | Status |
|---|---|---|---|
| 0 | 27 May 2026 | First issue (wheels only). Summary findings with figures. | Superseded |
| 1 | 08 Jun 2026 | Full hand calculations added for every check; methodology and assumptions expanded for third-party review. | Superseded |
| 2 | 09 Jun 2026 | Wheels-only detailed report. Cylinder-pin cross-check against Base Detail-4; column plastic-capacity check added. | Superseded by Rev 3 |
| A | 10 Jun 2026 | Deployed-outrigger study. Stability in the # arrangement with the legs at 90 degrees, plus the deployed leg-member check. Items above the slewing bearing carried from Rev 2. | Superseded by Rev 3 |
| 3 | 10 Jun 2026 | Combined wheels-only and deployed-outrigger verification in one document, for submission. Content consolidated from Rev 2 and Rev A; results unchanged. | Issued for review |

## Basis and limitations

- Drawings as issued. Section sizes, member lengths, weld throats, bolt grade, the slewing-bolt pattern and the leg dimensions are read directly off the supplied drawings. Nothing has been scaled by eye where a dimension callout exists.
- Self-weight. The crane mass of 700 kg is the figure supplied by the client. No bill of materials was available, so the mass split and centre-of-gravity offset are engineering estimates and are flagged in Section 26.
- Deployed geometry. The legs are taken square to the rails at 90 degrees, as shown on the deployment schematic. The deployed foot positions are part-dimensioned on the drawings and are flagged in Section 26.
- Scope of physics. Static lifting only. Wind, seismic action, side pull and snatch loading are outside this scope. A dynamic factor of 1.25 covers normal controlled hoisting per HC1 duty.
- Not a design. This is a verification of an existing unit against its rated chart in both stances. It is not a design certificate and does not replace load testing or a competent-person examination before the crane is put to work.

# Contents

# 1. Executive Summary

The crane was checked against its rated load chart in both working stances. Fourteen structural and mechanical checks were carried out, plus stability in each stance, by two independent methods: closed-form hand calculation and an in-script plane-frame finite element model. The two agree closely. The central finding is consistent across both stances and is easy to act on.

**The crane does not hold the working-stress margins of a rated 1 tonne appliance, in either stance.** Of the fourteen structural and mechanical checks, eight pass and six fall short, and both stability checks also fall short. The shortfalls are not spread evenly; they cluster on one operating point and one load path. The governing case is not the heaviest lift but the 700 kg load held out at 2320 mm, which produces the largest moment on the structure, 19.91 kN.m. At that point the slewing column reaches its yield stress (factor of safety 0.99) and the fillet weld at its base is worse again, at 0.85. These two items are the real headline; everything else that falls short is either marginal or sits on the same load path.

## 1.1 Wheels-only configuration

Standing on its wheels, the crane will pick the loads up, but at the close-in 1400 mm position the margin on the steel is 1.17 against the 1.5 target, and at mid reach it is at the limit of the steel. It cannot slew safely on its wheels, because the narrow rail track is the whole footprint. The column and its base weld, the boom outer section, the outrigger weld, the pivot and cylinder pins, and forward stability all fall short of target.

## 1.2 Deployed-outrigger configuration

Deploying the four legs in the # arrangement is what lets the crane slew, by widening the lateral track. But the legs sit at 90 degrees to the rails, so they reach sideways and not forward. Forward tipping is therefore unchanged from the wheels-only case and still fails at mid reach (0.96). Sideways tipping, the case the legs are there for, clears the target only close in; at mid and far reach it falls to 1.05 and 1.31 on the as-drawn 2376 mm track, and needs the track opened to about 2784 mm. The deployed leg member itself is comfortable at 1.91. Crucially, nothing above the slewing bearing changes: the column stays at 0.99 and its base weld at 0.85 in both stances.

## 1.3 Headline results

Governing load case, 700 kg at 2320 mm. Stability values are the worst across the chart in each stance.

| Item | Result | FoS | Target | Verdict |
|---|---|---|---|---|
| Column, combined N+M (both stances) | 277 MPa | 0.99 | 1.50 | **Fail** |
| Column-to-base weld (both stances) | 235 MPa | 0.85 | 1.50 | **Fail** |
| Boom outer section (both stances) | 186 MPa | 1.48 | 1.50 | **Marginal** |
| Outrigger arm bending (wheels) | 180 MPa | 1.53 | 1.50 | **Pass** |
| Cylinder rod-end pin (Ø20 as modelled) | 92 MPa | 1.20 | 3.00 | **Fail** |
| Forward stability (both stances) | 9.20 kN.m | 0.96 | 1.50 | **Fail** |
| Sideways stability, slewing (2376 track) | 7.77 kN.m | 1.05 | 1.50 | **Fail** |
| Deployed leg member (RHS 60x100x6) | 144 MPa | 1.91 | 1.50 | **Pass** |
| Column buckling, Pcr | 720 kN | 67.6 | 1.50 | **Pass** |
| Slewing bolts, utilisation 0.35 | — | 1.69 | 1.50 | **Pass** |

**Plastic reserve.** The column reaches first-fibre yield near the rated load, but as a compact section it holds reserve beyond first yield before a plastic hinge can form, which the elastic check does not credit; Section 17 sets this out. The factor of 0.99 is therefore a first-yield working-stress result rather than a collapse prediction, but the structure still does not hold the conventional 1.5 working-stress margin for sustained duty.

**Way forward.** The crane cannot be certified to the full 1 tonne chart on a working-stress basis in either stance. The recommended action is to re-rate the chart to what the steel safely carries, which needs no fabrication; the separate Load Re-Rating Report (Rev R1) sets a maximum SWL of 650 kg with reduced values at reach. Retaining the full 1 tonne rating requires the larger column section and matching base weld. Deploying the legs is a precondition for slewing, not a substitute for that structural work.

# PART I — Common basis (applies to both configurations)

## 2. Introduction and Scope

### 2.1 Background

The unit is a mobile floor jib crane built to lift up to 1 tonne. It runs on four wheels and carries a counterweight at the rear. A telescopic boom in three stages luffs on a single hydraulic cylinder, and the whole upper assembly slews on a bolted bearing flange above the base frame. The rated load chart marks three points: 1000 kg at 1400 mm, 700 kg at 2320 mm and 300 kg at 3300 mm. The warning plate instructs the operator to deploy the legs before slewing, so the rated chart is intended for the legs-down state.

The client commissioned an independent check to confirm whether the structure carries that chart with adequate margin against yielding, buckling, weld and pin failure, and overturning, in both stances. The work was scoped as a calculation exercise; no 3D model was built and no commercial finite element package was used.

### 2.2 What this report covers

The verification addresses, for the wheels-only stance, eight areas each reported with a factor of safety and a pass or fail against target: boom bending and tip deflection at each telescopic section; column stress under combined axial and bending plus buckling; base-frame wheel reactions and outrigger-arm bending; pin shear and bearing at the boom pivot, cylinder rod-end and hook; fillet-weld strength at the column base, boom bracket and outrigger; the slewing-flange bolt check; stability at every chart point; and a combined statement on the 1 tonne safe working load. For the deployed stance it adds forward and sideways stability, the deployed leg-member check, and a statement of which wheels-only results carry over unchanged.

### 2.3 Configurations assessed

Both stances are assessed in full. Where a result is identical in both, it is derived once and noted as common. The column and base-weld findings, in particular, stand in either configuration, because they sit above the slewing bearing and are fixed by the boom and the lifted load.

## 3. Materials, Sections and Allowable Stresses

### 3.1 Material and target factors of safety

All structural members are hollow steel sections in grade S275, taken from drawing General Note 5. The yield stress is fy = 275 MPa. Plates are A36 or equivalent, bolts are grade 8.8, and fillet welds are 6 mm throat minimum per Note 4. The allowable stress for each check follows from the target factor of safety: members are checked against yield with a target of 1.5; pins and bolts in shear and bearing carry a higher target of 3.0, as is normal for lifting-appliance connections.

**Allowable stresses**

```
Member bending / direct stress (target FoS 1.5):
   sigma_allow = fy / 1.50 = 275 / 1.50 = 183.3 MPa
Pin / bolt shear (target FoS 3.0):
   tau_allow,pin = 0.6 x fy / 3.0 = 0.6 x 275 / 3.0 = 55.0 MPa
Fillet weld (simplified, on throat, as coded):
   tau_allow,weld = 0.7 x fu / 1.5   (fu = 430 MPa)
```

### 3.2 Section properties

Section properties for a rectangular hollow section come from the standard expressions, where B and H are the outer width and depth and t the uniform wall thickness. The worked example for the outer boom is shown; the same expressions applied to every member give the properties used throughout, which match the values printed by the analysis model.

```
Rectangular hollow section, worked example: outer boom 100 x 140 x 6
   A  = B*H - (B-2t)*(H-2t) = 140*100 - 128*88 = 2736 mm^2 ~ 27.4 cm^2
   Ix = [B*H^3 - (B-2t)*(H-2t)^3] / 12 ~ 748.8 cm^4
   Sx = Ix / (H/2) = 748.8 / 7.0 = 107.0 cm^3
```

| Member | Section (RHS) | A (cm2) | Ix (cm4) | Sx (cm3) |
|---|---|---|---|---|
| Boom outer (root) | 100 x 140 x 6 | 27.4 | 748.8 | 107.0 |
| Boom middle | 80 x 120 x 6 | 22.6 | 438.2 | 73.0 |
| Boom inner (tip) | 60 x 100 x 6 | 17.8 | 227.4 | 45.5 |
| Column | 80 x 120 x 6 | 22.6 | 438.2 | 73.0 |
| Base frame / outrigger / leg | 60 x 100 x 6 | 17.8 | 227.4 | 45.5 |

# 4. Methodology

Two independent methods were run against the same geometry so the results could be cross-checked. The first is closed-form hand calculation following Eurocode 3 and EN 13001. The second is a small two-dimensional frame solver written inside the model. Both read their inputs from one parametric block, so any change to a section size or weld throat flows through every check at once.

## 4.1 Track 1: closed-form hand calculation

Each member is reduced to its governing internal action and checked by hand. Bending uses the elastic section modulus, axial load uses the gross area, and the two combine linearly for the column. Pins are checked in double or single shear and in bearing. Welds are checked on the throat area. Stability is a moment balance about the tipping line.

## 4.2 Track 2: in-script frame finite element check

The crane is modelled as a plane frame of Euler-Bernoulli beam elements, each carrying a 6 by 6 stiffness matrix rotated into the global frame and assembled into the system stiffness. The front wheel is pinned and the rear is a roller. Solving the free degrees of freedom gives the nodal displacements, from which element moments and stresses follow. The frame is re-solved for all three chart points because the boom extends to a different length at each.

```
Why a side-view 2D frame needs a correction
A side elevation only sees one base beam, but the real crane has two
parallel side rails carrying the load down to the wheels. The outrigger
and base elements in the frame model therefore use twice the area and
twice the second moment of area, lumping both rails into one equivalent
beam. Without this the model would overstate base-frame stress by a
factor of two. Crane self-weight is applied at the column-base node.
```

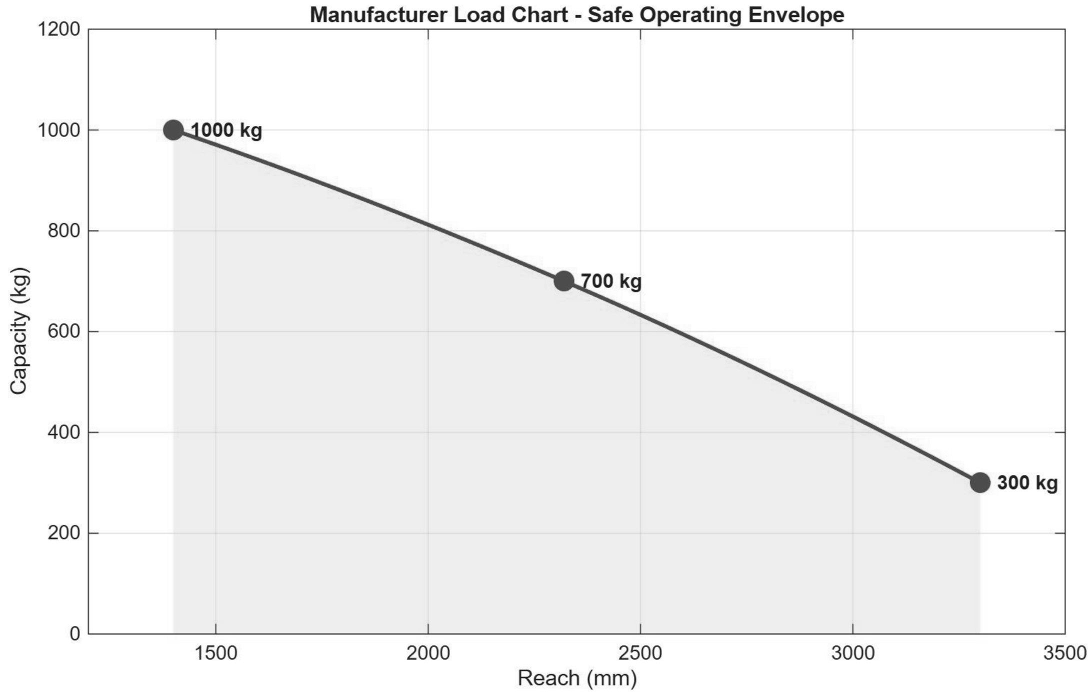
## 4.3 Cross-checking the two tracks

The two methods are expected to agree on the column-base moment and the peak member stress. Where they differ is informative: the hand calc treats the outrigger as a simple cantilever, while the frame model captures moment transfer through the bolted joint, and it adds column and base flexibility to the boom-tip movement. Section 16 reports the comparison.

# 5. Load Model and Governing Case

## 5.1 Rated load chart

The manufacturer chart gives three rated points; capacity falls as reach increases, the expected shape for a luffing jib. A dynamic amplification factor of 1.25 is applied to the static load, consistent with HC1 hoist duty under EN 13001.




*Rated load chart with the safe operating envelope shaded.*

| Chart point | Reach (mm) | Rated load (kg) | Static load (kN) | Factored load (kN) |
|---|---|---|---|---|
| Close | 1400 | 1000 | 9.81 | 12.26 |
| Mid | 2320 | 700 | 6.87 | 8.58 |
| Far | 3300 | 300 | 2.94 | 3.68 |

### 5.2 Which case governs

The design moment at the column is the factored load times its reach. Running the three points shows the mid-reach case is the most severe, even though it is not the heaviest load.

```
Design moment at column, M = W x r
   Close 1000 kg: M = 12.26 kN x 1.400 m = 17.17 kN.m
   Mid    700 kg : M =  8.58 kN x 2.320 m = 19.91 kN.m   <-- governs
   Far    300 kg : M =  3.68 kN x 3.300 m = 12.14 kN.m
```

Governing design moment: 19.91 kN.m at 700 kg / 2320 mm. The column, the base weld, the boom outer section and the stability check are all worst at this point. It is the case to keep in mind throughout the report.

## 6. Free Body and Design Forces

The factored tip load reacts through the boom into the luffing cylinder and the pivot, and from there down the column to the base and the wheels. The counterweight and crane self-weight act at the rear (free-body figure on the title page).

### 6.1 Cylinder and pivot forces

Taking moments about the boom pivot gives the cylinder force needed to hold the boom. The cylinder acts at a short lever arm relative to the load, so its force is large. The pivot then carries the vector sum of the cylinder force and the load reaction. The lever arm a_cyl is the perpendicular distance from the pivot to the cylinder line of action and is flagged for site confirmation in Section 26.

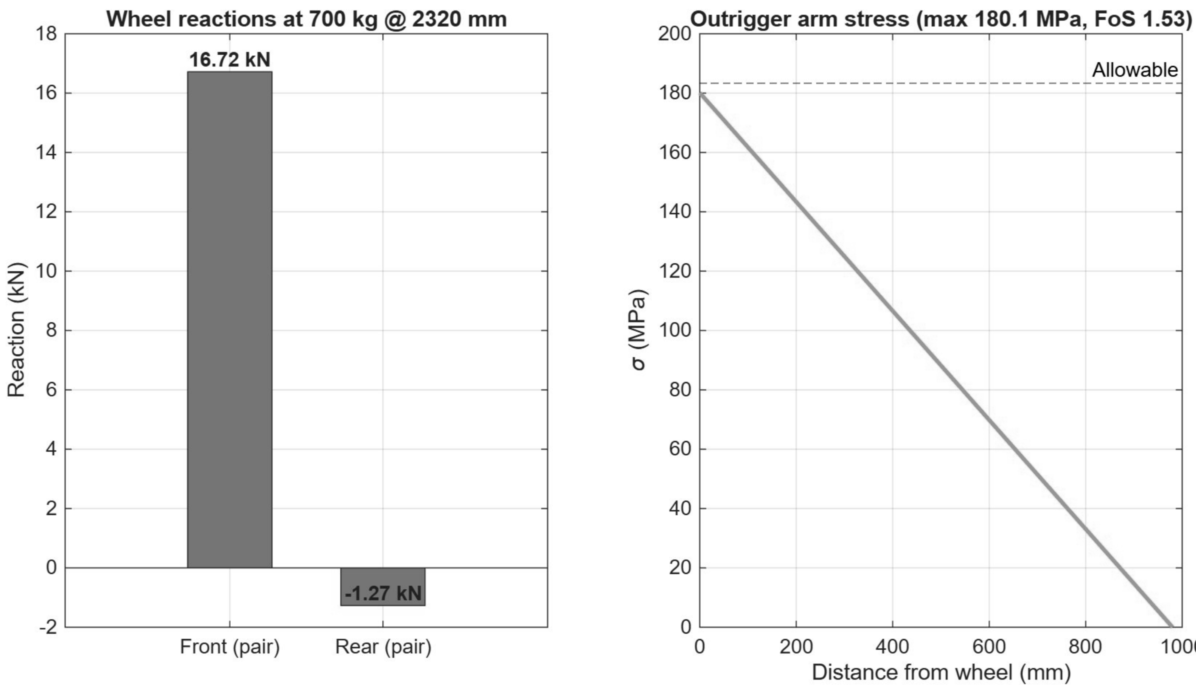
```
Boom moment balance about the pivot
   Cylinder force  F_cyl   = M_worst / a_cyl = 19.91 / 0.346 = 57.49 kN
   Pivot reaction  F_pivot = vector sum(F_cyl, W)            = 63.85 kN
```

```
Hydraulic pressure cross-check (luffing cyl = Type 80, bore 50 mm)
   Pressure required      = 57.49 kN / 1963 mm^2 = 29.3 MPa (293 bar)
   (a 30 mm bore would need 813 bar -> confirms 50 mm bore is the luffer)
```

## 6.2 Wheel reactions

Resolving the whole crane about the wheel lines gives the support reactions. Almost all of the load goes to the front wheel pair, with the rear pair barely loaded at the governing case, because the overturning moment has nearly lifted the rear off the ground. The near-zero rear reaction confirms the crane is close to tipping, which the stability check quantifies.

```
Vertical and moment balance on the base (worst case)
   R_front (pair) = 16.72 kN  -> 8.36 kN per wheel
   R_rear  (pair) = -1.27 kN  -> -0.63 kN per wheel
```




*Wheel reactions (left) and outrigger-arm stress profile (right) at the governing case.*

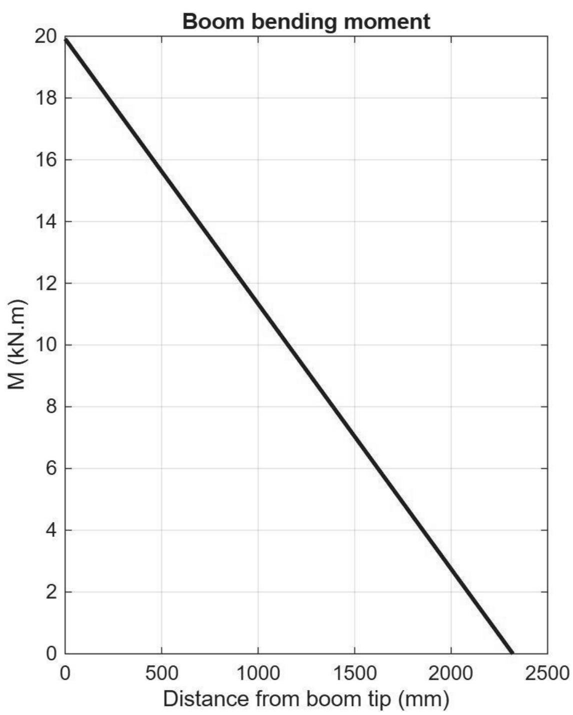
# PART II — Wheels-only configuration (outriggers stowed)

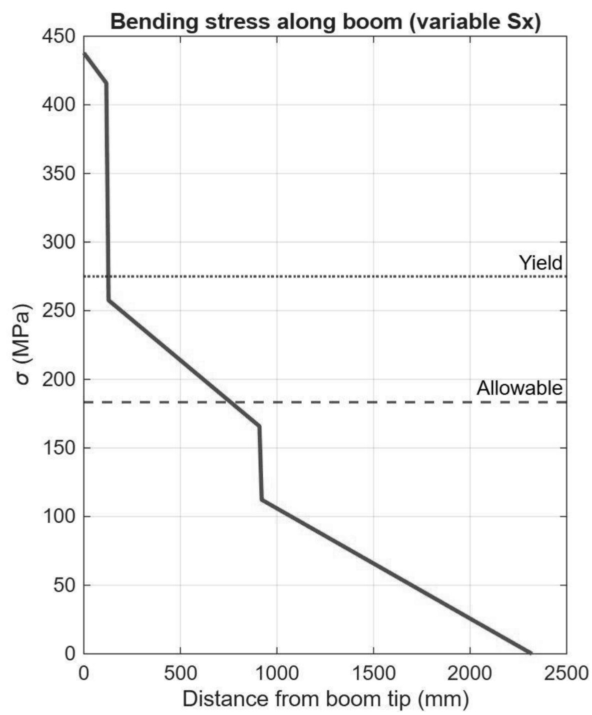
## 7. Boom Bending

The telescopic boom steps down in three sections from root to tip. Each section is checked at its own root, where its local moment is highest. The outer section at the column end carries the full design moment.

```
Boom outer (RSH 100 x 140 x 6) at root
   M = 19.91 kN.m   Sx = 107.0 cm^3
   sigma = 19.91e6 / 107.0e3 = 186.2 MPa   FoS = 275 / 186.2 = 1.48  (marginal)
Boom middle (RSH 80 x 120 x 6) at its root
   M = 7.90 kN.m    Sx = 73.0 cm^3
   sigma = 108.1 MPa   FoS = 2.54  (pass)
Boom inner (RSH 60 x 100 x 6) at its root
   M = 4.05 kN.m    Sx = 45.5 cm^3
   sigma = 89.0 MPa    FoS = 3.09  (pass)
```

Tip deflection at the worst case reaches about 23 mm by hand against an L/200 serviceability guide of roughly 12 mm, so the boom is outside the deflection guide at full reach. The frame model gives a larger tip movement because it includes column and base flexibility (Section 16).




*Boom bending moment (left) and stress along the boom with the stepped section (right). The outer section sits above the 183 MPa allowable.*

## 8. Column under Combined Load

The column carries axial load from the boom system plus the full bending moment transferred at its top. The two stresses add directly. This is the governing structural check, and it is identical in both stances.

```
Column (RSH 80 x 120 x 6), combined axial + bending
   N = 10.64 kN   A = 22.6 cm^2     M = 19.91 kN.m   Sx = 73.0 cm^3
   sigma_axial = 10.64e3 / 2260   =   4.7 MPa
   sigma_bend  = 19.91e6 / 73.0e3 = 272.7 MPa
   sigma_total =                    277.4 MPa
   FoS = 275 / 277.4 = 0.99  ->  Fail (at yield)

Column buckling (Euler, weak axis, K = 2 cantilever)
```

```
   lambda = 80.6   Pcr = 720 kN   N = 10.64 kN   FoS = 67.6  (pass)
```

The column is a bending problem, not a buckling problem. Its axial stress is tiny and buckling has a vast margin, but the bending stress alone uses up the entire yield capacity. To recover a 1.5 margin the section modulus would need to rise by about half, which points to a deeper or thicker column section rather than any change to axial capacity.

## 9. Base Frame and Outrigger Arm

The outrigger arm is checked as a cantilever picking up the front wheel reaction at its tip; the moment at the root gives the stress.

```
Outrigger arm (RSH 60 x 100 x 6)
   M_arm = 8.19 kN.m   Sx = 45.5 cm^3
   sigma = 8.19e6 / 45.5e3 = 180.1 MPa   FoS = 275 / 180.1 = 1.53  -> Pass
```

## 10. Pins and Connections

The pins are checked in shear (double or single as built) and in bearing on the lug. The cylinder force and pivot reaction from Section 6.1 drive the pin loads.

```
Boom pivot pin, dia 30 mm, double shear
   F_pivot = 63.85 kN   tau = 45.2 MPa   bearing = 141.9 MPa   FoS = 2.44  -> Fail*
Cylinder rod-end pin (as modelled, dia 20 mm), double shear
   F_cyl = 57.49 kN     tau = 91.5 MPa   bearing = 191.6 MPa   FoS = 1.20  -> Fail
Cross-check at Base Detail-4 pin (dia 40 mm), double shear
   tau = 22.9 MPa   bearing = 95.8 MPa   FoS = 4.81  -> Pass
Hook pin, dia 20 mm, single shear
   tau = 27.3 MPa   bearing = 28.6 MPa   FoS = 4.03  -> Pass
```

***** *On the conservative series-factor basis coded (material factor and pin factor applied in series), the pivot pin reads 2.44. On the conventional lifting-pin basis it moves to about 3.65 and passes. The basis-independent finding is that a 20 mm rod-end pin is undersized for the 57 kN it would carry, smaller even than the 30 mm pivot pin; the 40 mm Detail-4 pin clears the check on either basis. The summary table carries the conservative 20 mm figure. The as-built diameter should be confirmed on the machine.*

## 11. Welds

Fillet welds are checked on the throat area at 6 mm leg. The column-base weld carries the full moment as a stress couple across the weld group and is the governing weld; it governs alongside the column.

```
Column-to-base weld (6 mm fillet)   sigma = 234.9 MPa   FoS = 0.85  -> Fail
Boom-bracket weld    (6 mm fillet)   tau   =  35.5 MPa   FoS = 3.39  -> Pass
Outrigger weld       (6 mm fillet)   sigma = 155.0 MPa   FoS = 1.29  -> Fail
```

## 12. Slewing-Flange Bolts

The five M16 grade 8.8 bolts sit on a 170 mm pitch circle. The overturning moment is shared as a tension couple across the pattern using the polar-moment method, and combined tension-plus-shear is checked against the EN 1993-1-8 interaction.

```
5 x M16 grade 8.8 on 170 mm PCD, polar-moment method
   Bolt capacity : tension 113.0 kN, shear 75.4 kN
   Peak demand   : tension  93.7 kN, shear  2.1 kN per bolt
   Interaction utilisation = 0.35   ->   FoS (combined) = 1.69  -> Pass
```

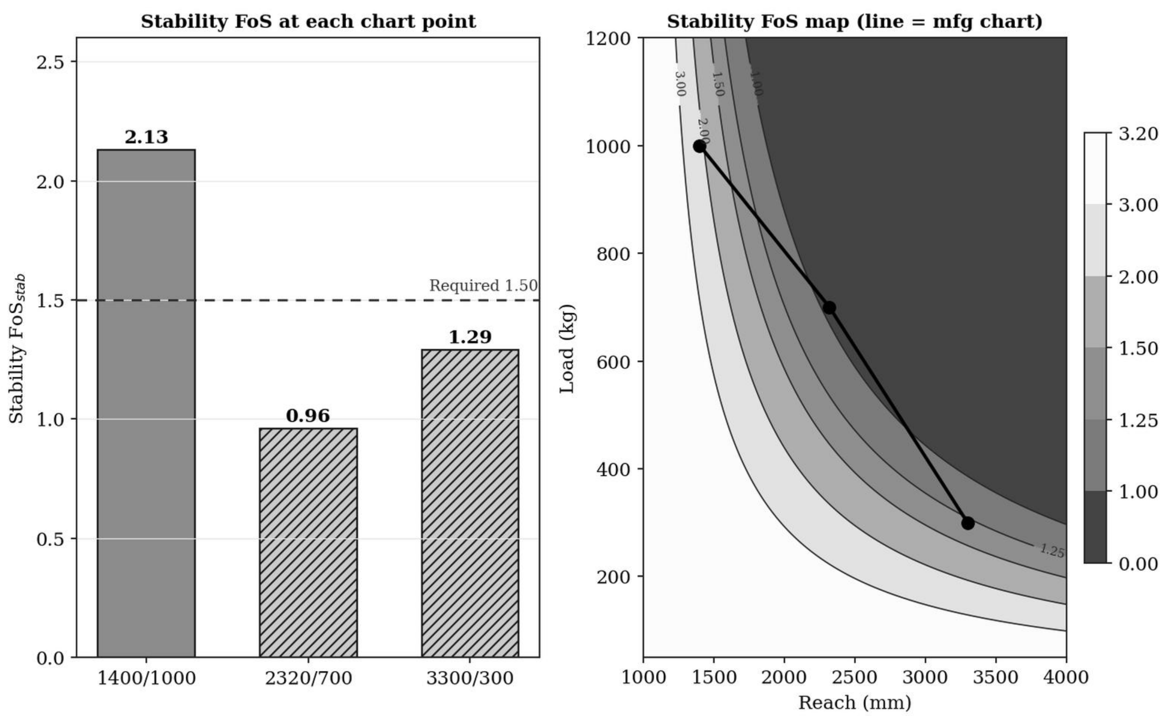
## 13. Stability against Overturning (wheels)

Stability is a moment balance about the front wheel line, 980 mm ahead of the column. The restoring moment comes from the crane self-weight and counterweight acting behind that line; the overturning moment from the load acting ahead of it. The mid-reach point governs.

```
Overturning balance about the front wheel line
   Restoring M_R = m_crane x g x (L_fwd + e_cg) = 700 x 9.81 x 1.280 = 8.79 kN.m
   Overturning M_OT = m_load x g x (reach - L_fwd)   [static load]
```

| Load case | M_OT (kN.m) | FoS = M_R / M_OT | Verdict |
|---|---|---|---|
| 1000 kg @ 1400 mm | 4.12 | 2.13 | **Pass** |
| 700 kg @ 2320 mm | 9.20 | 0.96 | **Fail** |
| 300 kg @ 3300 mm | 6.83 | 1.29 | **Fail** |

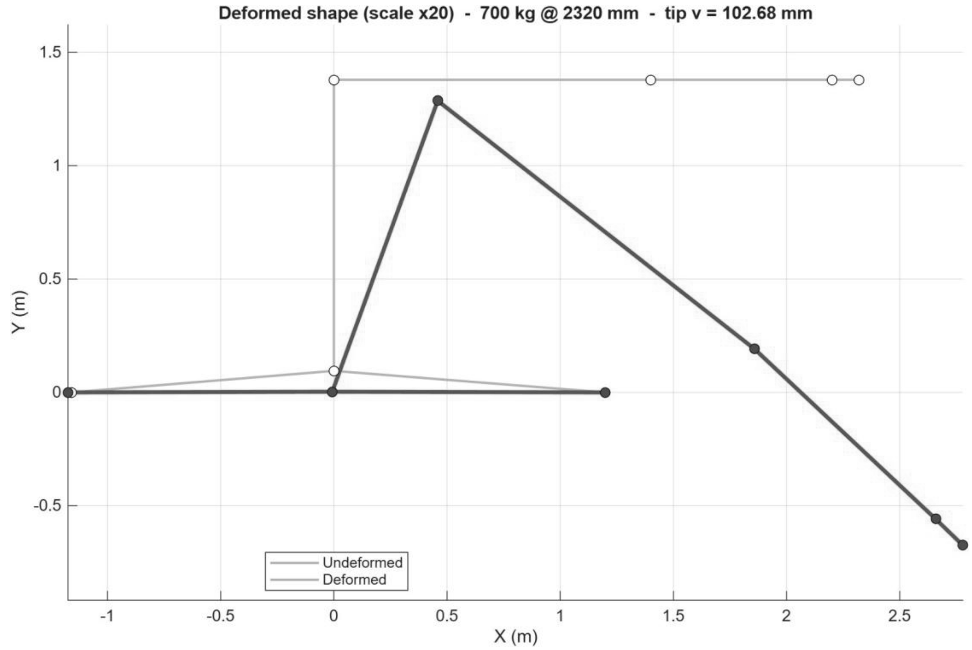
Worst stability factor 0.96 at 700 kg / 2320 mm against a target of 1.50, a fail in the wheels-only state. The near-zero rear wheel reaction of Section 6.2 is the same result seen from the support side.



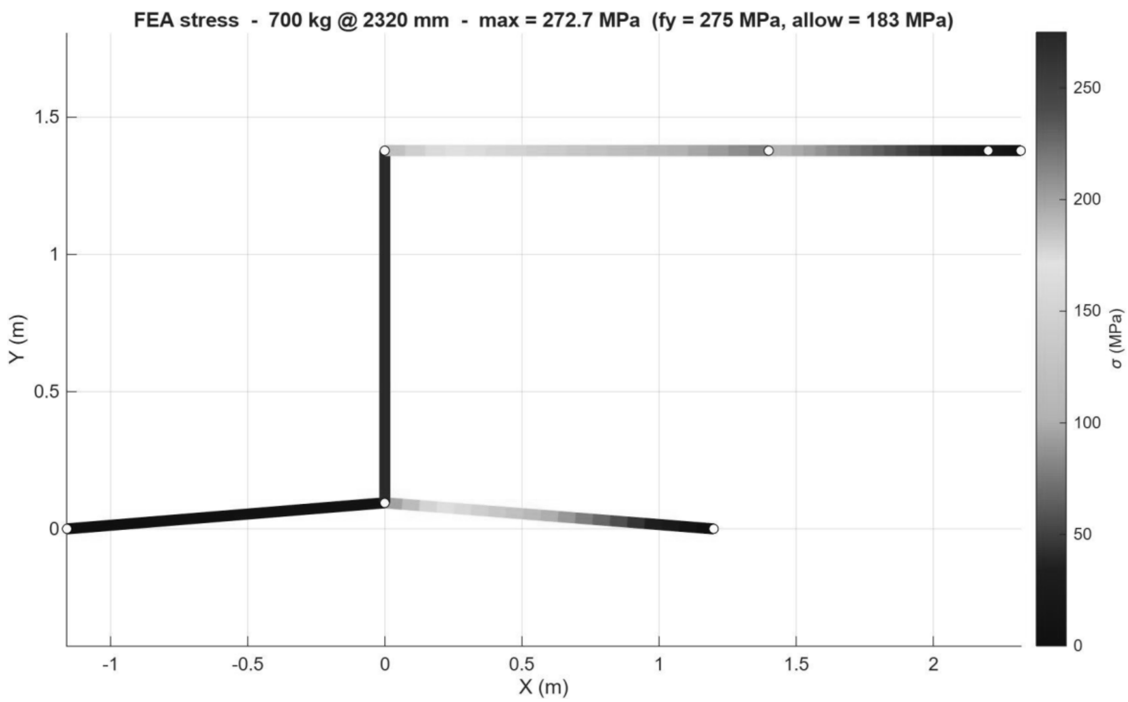

*Stability factor at each chart point (left) and the factor map across reach and load with the chart overlaid (right).*

## 14. Frame Model Results

The plane-frame solver was run for all three chart points. The deformed shape and the stress distribution for the governing case are shown below. The model confirms the column as the most-stressed member, in agreement with the hand calculation.




*Deformed shape at the governing case, scaled x20. Boom-tip movement reflects column and base flexibility, not boom bending alone.*



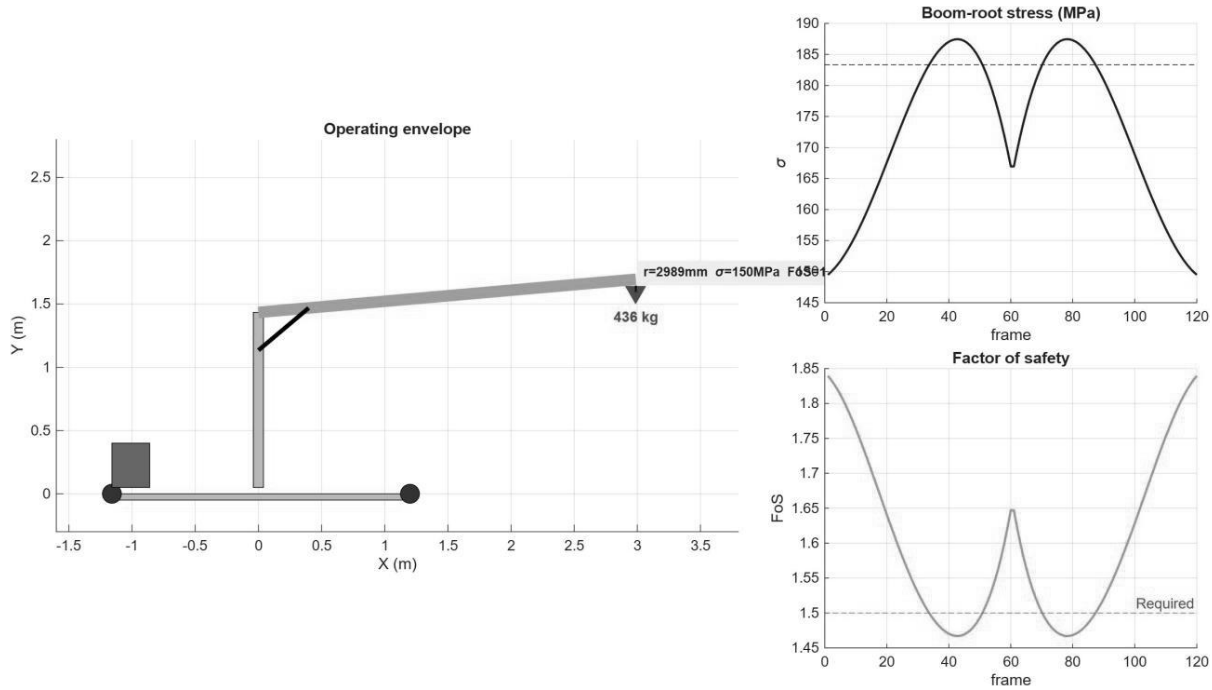

*Frame stress at the governing case. Peak 272.7 MPa at the column, against yield 275 MPa and the 183 MPa allowable.*

| Chart point | Tip deflection (mm) | Peak stress (MPa) | FoS |
|---|---|---|---|
| 1000 kg @ 1400 mm | 51.5 | 235.1 | 1.17 |
| 700 kg @ 2320 mm | 102.7 | 272.7 | 1.01 |
| 300 kg @ 3300 mm | 101.1 | 166.2 | 1.65 |

The frame FoS values are quoted against yield directly (275 / peak stress), which is why they read slightly higher than the 1.5-target factors in the hand calc. On a like-for-like basis the two methods agree, as Section 16 shows.

## 15. Operating Envelope

The model also sweeps the boom through its luffing range and tracks the boom-root stress and factor of safety against the chart capacity at each angle. The capacity drops at the reaches where stress approaches the allowable line.



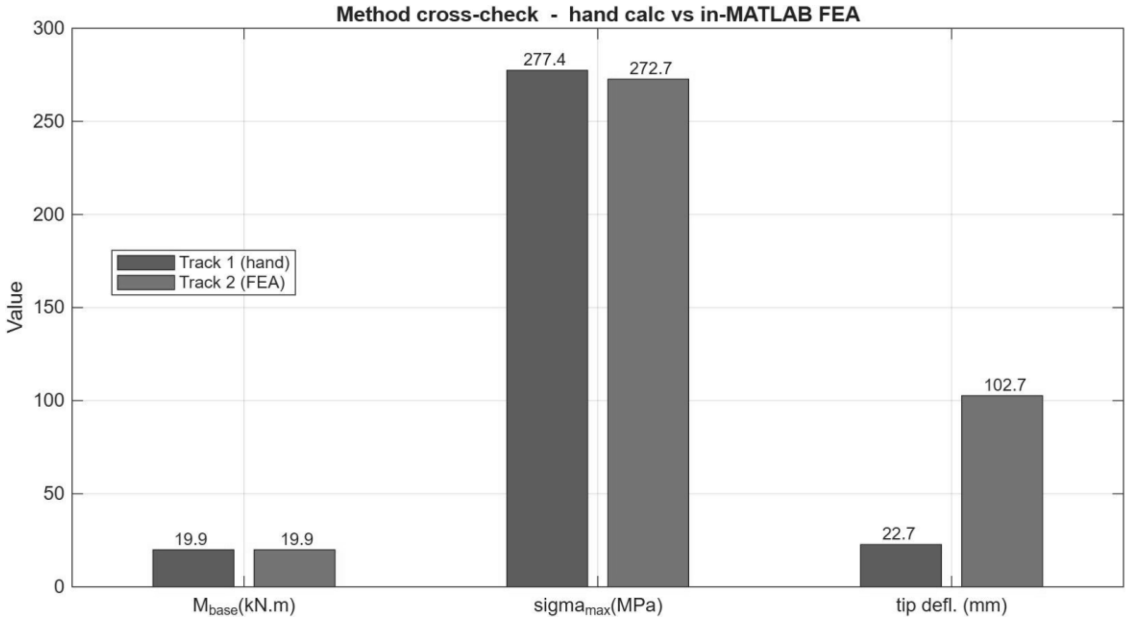

*Operating-envelope sweep. Left: crane pose. Right: boom-root stress and factor of safety through the luffing cycle, with the 183 MPa allowable and 1.5 target marked.*

## 16. Cross-Validation of the Two Methods

The hand calculation and the frame model are compared on the three quantities they both produce. The column-base moment matches exactly, the peak stress agrees to within two percent, and the tip deflection differs because the two measure different things.

| Quantity | Track 1 (hand) | Track 2 (frame) | Agreement |
|---|---|---|---|
| Column-base moment (kN.m) | 19.91 | 19.91 | Exact |
| Peak member stress (MPa) | 277.4 | 272.7 | Within 2% |
| Boom tip deflection (mm) | 22.7 | 102.7 | Different basis |

The deflection gap is expected and not an error. The hand figure is boom bending alone, treating the column top as rigid; the frame figure adds the column bending and the base flexibility, so the tip moves much further. For a stress verdict the moment and stress agreement is what matters, and that is tight.




*Side-by-side comparison of the two methods.*

## 17. Elastic and Plastic Capacity of the Column

The column reaches first-fibre yield at a working-stress factor of safety of 0.99, which means the applied moment at 1.25 times the rated load is essentially at the first-yield moment of the section. First yield, however, is not collapse. The collapse limit for a compact steel section is the plastic moment, which is larger than the first-yield moment by the shape factor, 1.23 for this hollow section. That difference is reserve the elastic working-stress check does not credit.

```
Column RSH 80 x 120 x 6, elastic-to-plastic capacity (nominal S275)
   Sx = 73.0 cm^3   Zx = 89.7 cm^3   shape factor Zx/Sx = 1.23
   applied moment @ 1.25 x SWL = 19.91 kN.m   (elastic FoS 0.99)
   first-yield moment My       = 20.08 kN.m
   plastic moment Mp (nom 275) = 24.67 kN.m   (reserve over My +23%)
   plastic moment Mp (mill 320)= 28.71 kN.m   (reserve over My +43%)
   plastic margin on applied moment: Mp / 19.91 = 1.24 (nominal S275)
```

On nominal S275 the plastic moment is 24.67 kN.m against the applied 19.91 kN.m, a plastic margin of about 1.24, and on realistic mill steel the margin is wider. Two further effects act in the same direction: the calculations carry a 1.25 dynamic factor, and the luffing cylinder acting as a diagonal strut would relieve some of the column bending the cantilever model does not credit. The other members stay well within the elastic range, which is why the column is the lone item near its limit. The factor of safety of 0.99 is a first-yield working-stress result, not a prediction of failure; what the structure does not have, as built, is the conventional 1.5 working-stress margin at mid reach for repeated service.

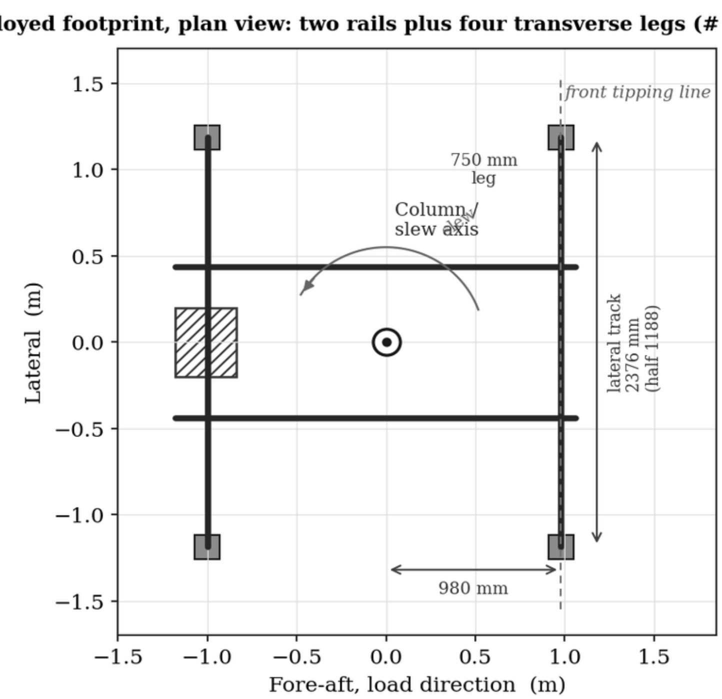
# PART III — Deployed-outrigger configuration (# arrangement)

## 18. The Deployed Configuration

The crane base is a pair of fore-aft rails set about 876 mm apart, with a wheel at each corner. Stowed, that narrow rail track is the whole footprint, and it is the reason the crane is unsafe to slew on its wheels. Deploying the legs adds two transverse beams, one near the front wheel line and one near the rear, each reaching to both sides. The four feet at their ends, with the two rails, make the # seen in plan.

The leg dimensions are read from Base DETAIL-11: each leg arm is 750 mm long, with a 230 mm drop to an 80 mm foot pad, an M16 jack screw for levelling, and a Ø32 hinge pin. Because the beams sit square to the rails, the feet widen the footprint sideways only. The front foot stays level with the front frame line, 980 mm ahead of the column, so it adds nothing to the forward overhang. The half-track is 438 plus 750, which is 1188 mm, and the full track is 2376 mm. This single number governs sideways stability.

| Quantity | Value | Source |
|---|---|---|
| Leg arm (lateral reach) | 750 mm | DETAIL-11 |
| Foot drop / pad | 230 mm / 80 mm | DETAIL-11 |
| Levelling jack screw | M16 (Ø16) | DETAIL-11 |
| Leg hinge pin | Ø32 | DETAIL-11 |
| Alternative leg arm | 798 mm | DETAIL B-B |
| Frame track, rail to rail | 876 mm | Base drawing |
| Deployed lateral track | 2376 mm (876 + 2 x 750) | derived |
| Front foot ahead of column | 980 mm (front frame line) | Base drawing |
| Leg member section | RHS 60 x 100 x 6 | drawing |




*Deployed footprint in plan. The legs reach 750 mm to each side, giving a 2376 mm track, while the front foot stays at the 980 mm front line.*

## 19. Forward Stability (deployed)

Forward tipping is a moment balance about the front foot line, 980 mm ahead of the column. With the legs at 90 degrees, that line is in the same place as the front wheel, so the balance is identical to the wheels-only case. Transverse legs do not move the front line, so they cannot improve forward stability.

| Load case | M_OT (kN.m) | FoS = M_R / M_OT | Verdict |
|---|---|---|---|
| 1000 kg @ 1400 mm | 4.12 | 2.13 | **Pass** |
| 700 kg @ 2320 mm | 9.20 | 0.96 | **Fail** |
| 300 kg @ 3300 mm | 6.83 | 1.29 | **Fail** |

The mid-reach point governs at 0.96, below target, matching the wheels-only result as it must.

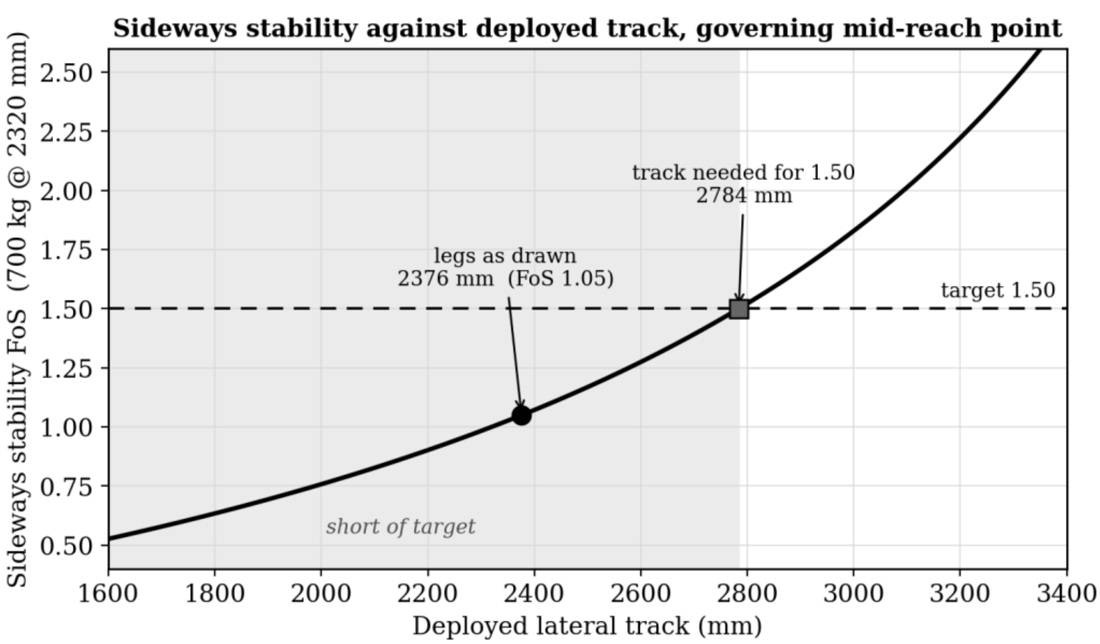
## 20. Sideways Stability During Slew

Sideways tipping is the case the legs are there for. With the boom slewed across the rails, the crane can tip over a side foot line set at the half-track, 1188 mm from the centreline. The restoring moment is the crane weight at that half-track; the overturning moment is the lifted load at its reach less the half-track.

```
Overturning balance about a side foot line
   Restoring M_R = m_crane x g x t_half = 700 x 9.81 x 1.188 = 8.16 kN.m
   Overturning M_OT = m_load x g x (reach - t_half)   [static load]
   Half-track from the legs: t_half = 438 + 750 = 1188 mm (track 2376 mm)
```

| Load case | M_OT (kN.m) | FoS = M_R / M_OT | Verdict |
|---|---|---|---|
| 1000 kg @ 1400 mm | 2.08 | 3.92 | **Pass** |
| 700 kg @ 2320 mm | 7.77 | 1.05 | **Fail** |
| 300 kg @ 3300 mm | 6.22 | 1.31 | **Fail** |

The close-in lift is steady, but the mid and far points fall below target, at 1.05 and 1.31. Solving the balance for 1.50 at the governing mid point gives a half-track of 1392 mm, a full track of 2784 mm. The legs as drawn give 2376 mm, about 400 mm short. The manufacturer figure marks the deployed width as adjustable up to roughly 2870 mm, where the worst sideways factor lifts to about 1.6. The deployed foot positions must therefore be measured on the machine and set to at least 2784 mm before slewing at mid or long reach.




*Sideways stability against the deployed track at the governing mid-reach point. The 2376 mm track sits below the 1.50 target; about 2784 mm is needed.*

## 21. The Deployed Leg Member

The deployed leg is checked as a cantilever picking up the vertical reaction at its foot. The largest foot reaction comes from the heaviest lift taken in close, not from the tipping case, because member force follows the vertical load while tipping follows the lever arm. At 1000 kg and 1400 mm the front reaction is 17.43 kN, shared by the two front feet.

```
Deployed leg, RHS 60 x 100 x 6, cantilever
   front reaction (1000 kg @ 1400 mm) = 17.43 kN total -> 8.72 kN per leg
   lever (leg arm) = 0.750 m
   M_leg = 8.72 x 0.750 = 6.54 kN.m   Sx = 45.5 cm^3
   sigma = 6.54e6 / 45.5e3 = 143.7 MPa   FoS = 275 / 143.7 = 1.91  -> Pass
```

The leg passes at 1.91. It is less stressed than the wheels-only outrigger arm, which carries the front reaction over the longer 980 mm fore-aft cantilever and runs at 180 MPa for 1.53. The base frame between the column and the front legs still carries that front reaction much as it does to the wheels, so the Rev 2 outrigger-arm result stays representative for that member.

## 22. Items Carried Over from the Wheels-Only Check

The legs attach to the base frame, below the slewing bearing. The axial load and the bending moment that pass up through the bearing into the column, the boom and its pins are fixed by the boom geometry and the lifted load, and do not change when the base stands on legs instead of wheels. The wheels-only results for those members therefore apply without alteration.

| Check | FoS | Target | Verdict |
|---|---|---|---|
| Boom outer (100x140x6) | 1.48 | 1.50 | **Marginal** |
| Boom middle (80x120x6) | 2.54 | 1.50 | **Pass** |
| Boom inner (60x100x6) | 3.09 | 1.50 | **Pass** |
| Column, combined N+M | 0.99 | 1.50 | **Fail** |
| Column buckling | 67.6 | 1.50 | **Pass** |
| Base outrigger arm (fore-aft cantilever) | 1.53 | 1.50 | **Pass** |
| Pivot pin (Ø30) | 2.44 | 3.00 | **Fail** |
| Cylinder rod-end pin (Ø20 as modelled) | 1.20 | 3.00 | **Fail** |
| Hook pin (Ø20) | 4.03 | 3.00 | **Pass** |
| Slewing bolts (5 x M16, 8.8) | 1.69 | 1.50 | **Pass** |
| Weld, column base | 0.85 | 1.50 | **Fail** |
| Weld, boom bracket | 3.39 | 1.50 | **Pass** |

The column at 0.99 and its base weld at 0.85 remain the governing structural items in both configurations.

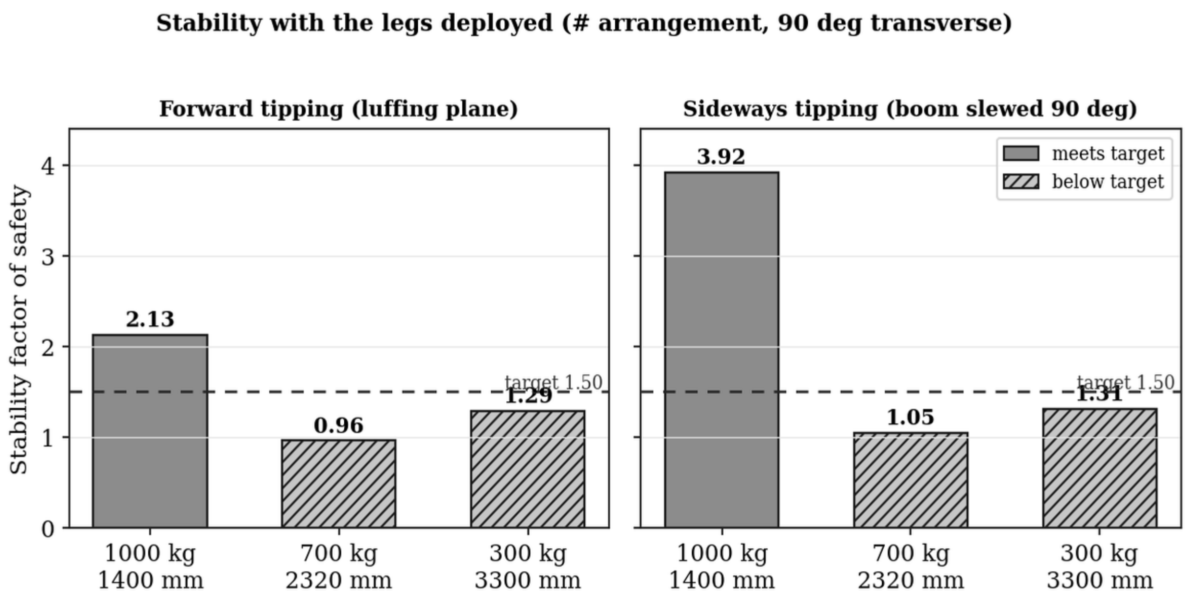
## 23. Stowed versus Deployed

The legs change the bottom of the load path and the stability picture; they leave the top of the structure exactly as it was.

| Item | Wheels stowed | Legs deployed (# arrangement) |
|---|---|---|
| Front tipping line ahead of column | 980 mm | 980 mm (no forward reach) |
| Lateral foot track | 876 mm | 2376 mm |
| Forward tipping, worst point | FoS 0.96, Fail | FoS 0.96, Fail (unchanged) |
| Sideways tipping during slew | narrow track, tips | FoS 1.05, Fail (track short) |

| Item | Wheels stowed | Legs deployed (# arrangement) |
|---|---|---|
| Front ground-support member | outrigger arm 180 MPa, FoS 1.53, Pass | leg arm 144 MPa, FoS 1.91, Pass |
| Column, combined N+M | 277 MPa, FoS 0.99, Fail | unchanged |
| Column-to-base weld | 235 MPa, FoS 0.85, Fail | unchanged |
| Boom outer section | 186 MPa, FoS 1.48, Marginal | unchanged |
| Pins and slewing bolts | as Rev 2 | unchanged |

Read across, the pattern is consistent: the legs widen the track and ease the front-support member, so sideways tipping becomes possible to manage and the leg member passes; forward tipping and the column, weld and boom are the same in both states, because the legs do not act on them.



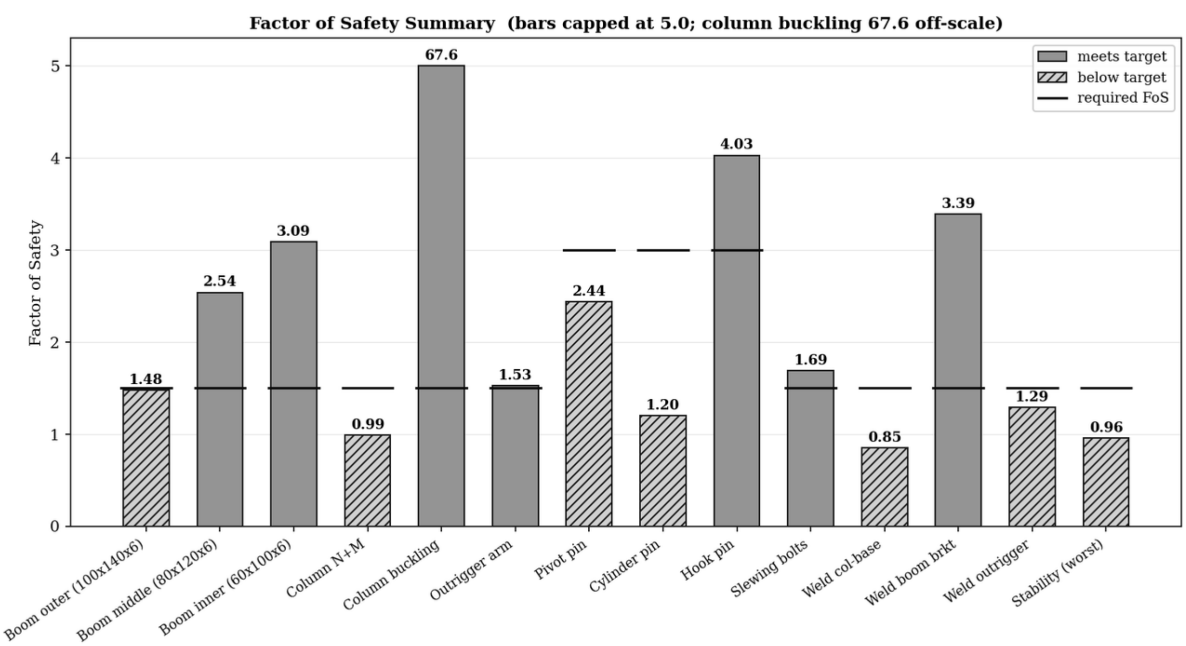

*Stability in the deployed state. Forward tipping is unchanged from the wheels-only case; sideways tipping clears the target only close in.*

# PART IV — Combined outcome

## 24. Summary of Results

All sixteen checks are collected below with their factors of safety and verdicts against target. Eight of the sixteen do not meet target on a working-stress basis. The column and its base weld govern; the boom outer section is a borderline miss. Stability is shown for both stances.

| Check | FoS | Target | Verdict |
|---|---|---|---|
| Boom outer (100x140x6) | 1.48 | 1.50 | **Marginal** |
| Boom middle (80x120x6) | 2.54 | 1.50 | **Pass** |
| Boom inner (60x100x6) | 3.09 | 1.50 | **Pass** |
| Column, combined N+M | 0.99 | 1.50 | **Fail** |
| Column buckling | 67.62 | 1.50 | **Pass** |
| Outrigger arm | 1.53 | 1.50 | **Pass** |
| Pivot pin | 2.44 | 3.00 | **Fail** |
| Cylinder pin (Ø20 as modelled) | 1.20 | 3.00 | **Fail** |
| Hook pin | 4.03 | 3.00 | **Pass** |
| Slewing bolts | 1.69 | 1.50 | **Pass** |
| Weld, column base | 0.85 | 1.50 | **Fail** |
| Weld, boom bracket | 3.39 | 1.50 | **Pass** |
| Weld, outrigger | 1.29 | 1.50 | **Fail** |
| Stability, forward (both stances) | 0.96 | 1.50 | **Fail** |
| Stability, sideways slew (2376 track) | 1.05 | 1.50 | **Fail** |
| Deployed leg member (RHS 60x100x6) | 1.91 | 1.50 | **Pass** |




*Factor of safety for every wheels-only check against the 1.5 / 3.0 targets.*

## 25. Conclusions and Recommendations

### 25.1 Conclusion on the 1 tonne SWL

In neither stance does the crane meet the agreed factors of safety across its rated chart. On its wheels, the steel carries the 1 tonne load in close only to a factor of 1.17, and at the governing 700 kg mid-reach point the column reaches yield (0.99) and its base weld is past target (0.85). Deploying the legs makes slewing possible and eases the front-support member, but it does not touch the column, the weld or forward tipping, all of which sit above the slewing bearing. The crane therefore cannot be certified to the full 1 tonne chart on a working-stress basis in either configuration.

Put plainly for the operator: the crane will lift the loads, but it does not carry the working-stress margin a rated 1 tonne crane is expected to have, and at medium reach it is at the limit of the steel. The lighter mid-reach lift is the more demanding one, not the heavy close lift, because of the longer lever arm.

### 25.2 The governing items

- Column, FoS 0.99 (both stances). At yield in bending; the clear primary item, though it retains plastic reserve as Section 17 shows.
- Column-base weld, FoS 0.85 (both stances). Past target on the same load path; addressed alongside the column.
- Sideways track, 2376 mm against 2784 mm needed. The one item the deployed configuration itself must fix; confirm the deployed width and, if the feet adjust, set the track to at least 2784 mm before slewing.
- Cylinder rod-end pin, FoS 1.20 (20 mm). Undersized if the pin is 20 mm; if it is the 40 mm Detail-4 pin it passes at 4.81. Confirm the as-built size.
- Boom outer section, FoS 1.48. A marginal miss at the 1.50 line that the recommended re-rate clears. The outrigger arm now sits just above target at 1.53 on the corrected geometry.

### 25.3 Recommendations

The recommended action and the route to retaining full capacity are set out below; the reduced chart itself is given in the separate Load Re-Rating Report (Rev R1).

| Route | Outcome |
|---|---|
| **Re-rate the chart (recommended)** | Reduce the rated capacities so every working-stress check clears target. The separate Re-Rating Report sets a maximum SWL of 650 kg, with about 390 kg at mid reach and 255 kg at far reach. Cheapest route, no fabrication, applies in both stances. |
| **Strengthen the column and base (keeps full rating)** | A larger column section, of the order of 100 x 150 x 8 in S275, with the base weld grown to suit and the cylinder pin confirmed or upsized. The only route that preserves the full 1 tonne chart. |
| **Confirm mill yield, cylinder geometry and deployed track** | Parallel actions. Confirming actual mill yield and the cylinder anchor would likely lift the column margin above 1.0; confirming the deployed track to 2784 mm clears sideways stability. |

**Recommendation.** Adopt the re-rate as the immediate action so the crane can be used safely at a reduced, verified rating, and hold the column upgrade as the path to retaining the full 1 tonne chart. Deploy the legs and confirm the deployed track to at least 2784 mm before any slew. The crane should not be worked to the current 1 tonne chart until the column, base weld and cylinder pin are addressed.

## 26. Assumptions and Items to Confirm

The section sizes, lengths, material grade, weld throats, bolt grade, slewing pattern and leg parts are taken directly from the drawings and are not in doubt. The following are estimates or part-dimensioned and should be confirmed against the built machine before the report is relied on for certification.

- Crane self-weight, 700 kg. Client-supplied with no bill of materials. A swing of plus or minus 100 kg moves both stability results.

- Centre-of-gravity offset, 0.30 m behind the column. Estimated from the counterweight and pump positions; affects stability only.
- Deployed lateral track, 2376 mm, and leg deployment angle, 90 degrees. Derived from the 876 mm frame track and the 750 mm DETAIL-11 leg arm. The make-or-break input for sideways stability; the manufacturer figure suggests up to about 2870 mm. The product photograph appears to show the front legs splayed forward, which would improve forward stability; the as-built angle and width must be measured.
- Front overhang to the tipping line, 980 mm. Used for stability and wheel reactions. The warning-plate 420 mm dimension should be reconciled against this on the machine.
- Cylinder lever arm (0.346 m) and rod-end pin diameter. The cylinder anchor geometry reproduces the issued cylinder force but is not dimensioned; the rod-end pin diameter is to be confirmed (Sections 10 and 17).
- Material yield. Nominal S275 (275 MPa) is used. Mill certificates or a hardness check would confirm the actual yield, which Section 17 shows is likely higher and would recover column margin.
- Weld throats, 6 mm, the drawing minimum. If site welds are smaller the weld checks worsen.
- Pin verification basis. The reviewer should confirm whether the conservative series-factor basis or the conventional lifting-pin basis applies to the pin verdicts (Section 10).

## 27. Standards Referenced

- EN 13001-3-1, crane structural design, load effects and proof of competence
- EN 1993-1-1, design of steel structures, general rules
- EN 1993-1-8, design of joints
- Cross-referenced with AS 1418.1 and SS 559 for the Singapore context

### Appendix A: Companion files

- Load Re-Rating Report (Rev R1), which converts these findings into a reduced SWL chart with operating limitations.
- MATLAB R2026a parametric model (project 40467361), the single file driving every number in this report, with the deployed case added as Test 10.
- Console log of the full run, including the factor-of-safety tables, the deployed stability tables, the leg-member check and the column plastic-capacity check.
- Figures as used in this report, and the workspace file for re-plotting without re-running.